| Name: | Class: | Date given: |
|---|---|---|
| | | Date due in: |

# Pressure



**A**



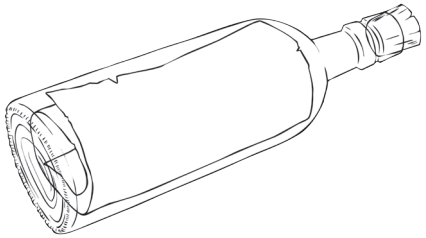
**B**



**C**

1a. Which shoe exerts the most pressure on to the floor? [1]

1b. Which shoe exerts the least pressure on to the floor? [1]



2. Meena is at the seaside and drops a message in a bottle into the sea. Explain why the bottle floats and will not sink. [2]

**pressure ($N/m^2$) = force (N) / area ($m^2$)**

3a. A man laid on a pool lilo has a surface area of $2m^2$ and a weight of 900N. Calculate the pressure. [2]

_____$N/m^2$



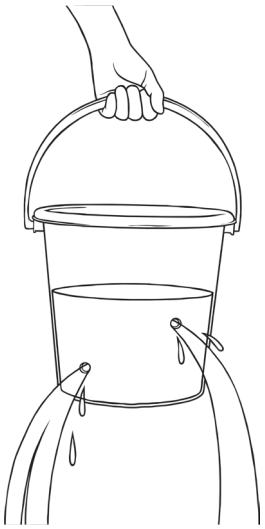
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3b. A horse weighs 2400N and each hoof has a surface area of 6cm$^2$. Calculate the pressure when the horse rears onto two legs? [2]

_____N/m$^2$

3c. A dog weighs 300N and each paw has a surface area of 5cm$^2$. Calculate the pressure when the dog is standing on all four paws? [2]

_____N/m$^2$



4. Milo has filled a bucket with water. He notices that there are two holes in the bucket where water is leaking. He observes how the water leaks from the holes at different speeds and angles. Explain why the water leaks differently from each hole. [3]

**Learning Outcomes** (tick if achieved)

| | | |
|---|---|---|
| Q1 | I can identify objects exerting more or less pressure | |
| Q2 | I can describe the effect of pressure in fluids | |
| Q3 | I can calculate pressure using an equation | |
| Q4 | I can explain differences in water pressure | |



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# Pressure **Answers**

1a. Which shoe exerts the most pressure on to the floor? [1] **A**

1b. Which shoe exerts the least pressure on to the floor? [1] **B**

2. Meena is at the seaside and drops a message in a bottle into the sea. Explain why the bottle floats and will not sink. [2]

**upward pressure exerted by the water is balanced with downward pressure exerted by the weight of the air**

3a. A man laid on a pool lilo has a surface area of $2m^2$ and a weight of 900N. Calculate the pressure. [2]

**$900N \div 2m^2 = 450\ N/m^2$**

3b. A horse weighs 2400N and each hoof has a surface area of $6cm^2$. Calculate the pressure when the horse rears onto two legs? [2]

**($6cm^2 \rightarrow 0.0006m^2 \times 2 = 0.0012m^2$)**

**$2400N \div 0.0012m^2 = 2{,}000{,}000\ N/m^2$ or $200N/cm^2$**

3c. If a dog weighs 300N and each paw has a surface area of $5cm^2$. Calculate the pressure when the dog is standing on all four paws? [2]

**($5cm^2 \rightarrow 0.0005m^2 \times 4 = 0.002m^2$)**

**$300N \div 0.002m^2 = 150{,}000\ N/m^2$ or $15N/cm^2$**

4. Milo has filled a bucket with water. He notices that there are two holes in the bucket where water is leaking. He observes how the water leaks from the holes at different speeds and angles. Explain why the water leaks differently from each hole. [3]

- **pressure increases with depth**
- **more weight from increased volume of water above**
- **therefore more pressure exerted downward**
- **increased pressure forces water out at greater speed**